

Exercise1

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Exercise 1

A shooter hits the target in a single shot with the probability of 0.95. Find the probability that in the next two shots the shooter succeeds to hit the target

- a. twice,
- b. at least once,
- c. exactly once.

(Ans. a. 0.9025, b. 0.9975, 0.095)

a. twice (independent events) $P = 0.95 \times 0.95 = 0.9025$

b. at least once (complement rule)

1 minus the probability of missing both

$$P = 1 - (0.05)^2 = 0.9975$$

c. exactly once

Case 1: first shot succeeds, second fails, 0.95×0.05

Case 2: first shot fails, second succeeds, 0.05×0.95

Add them

$$P = (0.95 \times 0.05) + (0.05 \times 0.95) = 0.095$$